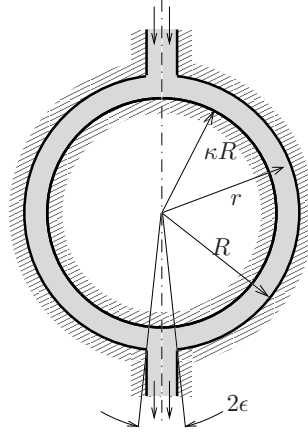


PROBLEM 1 (90 minutes)

A very viscous liquid (of constant density ρ and viscosity μ) flows steadily through the gap between two concentric spheres, as shown in the figure. The flow is driven by a given pressure difference between the entrance and exit section, $\Delta p = p_i - p_o$, and gravity effects are negligible. We wish to determine the resulting volume flux Q as a function of the known geometry data, fluid properties and pressure difference, $Q(R, \kappa, \epsilon, \rho, \mu, \Delta p)$.



1. Apply the dimensional analysis technique to write the parametric dependence of Q as a function of the minimum number of parameters.
2. Simplify the non-dimensional relation for the case of flow dominated by viscous forces, for which the liquid inertia is negligible.
3. Disregarding end effects in the entrance and exit regions, we can assume that the flow is unidirectional and symmetric, $\bar{v} = v_\theta(r, \theta)\bar{e}_\theta$ (note that $v_r = v_\phi = 0$ and $\partial/\partial\phi = 0$). With this assumption, write the simplified continuity and momentum equations in spherical coordinates, as well as the necessary boundary conditions to solve the system. Write the expression to evaluate the volume flux Q and then check that the parametric dependence stated earlier, that is $Q(R, \kappa, \epsilon, \rho, \mu, \Delta p)$, is correct.
4. Simplify the θ -momentum equation with the assumption that the velocity is small enough for the viscous terms to be dominant over the convective terms. Under this conditions, the radial pressure variations are negligible compared with Δp and we can consider that the pressure is a function of θ alone, $p(\theta)$.
5. Write the mathematical formulation (equations and boundary conditions) in non-dimensional form. Use the non-dimensional variables $\tilde{r} = r/R$, $\tilde{v}_\theta = v_\theta/V_\theta$ and $\tilde{p} = (p - p_o)/\Delta p$, where you can define V_θ such that there are no parameters in the non-dimensional equation.
6. Using the continuity equation, show that $\tilde{v}_\theta \sin \theta = \tilde{u}(\tilde{r})$, where $\tilde{u}(\tilde{r})$ is a function of $\tilde{r}$ to be determined.
7. Rewrite the θ -momentum equation in terms of the function $\tilde{u}(\tilde{r})$ instead of $\tilde{v}_\theta$ and separate variables to show that

$$\sin \theta \frac{d\tilde{p}}{d\theta} = C$$

$$\frac{1}{\tilde{r}} \frac{d}{d\tilde{r}} \left(\tilde{r}^2 \frac{d\tilde{u}}{d\tilde{r}} \right) = C$$

where C is a constant.

8. Integrate the ODE for the pressure and determine the value of the constant C . Note that the boundary conditions for the pressure are to be applied at $\theta = \epsilon$ and $\theta = \pi - \epsilon$.
9. With the known value of C , integrate the ode for $\tilde{u}(\tilde{r})$ obtained in item 7 and thus obtain the velocity profile $\tilde{v}_\theta(\tilde{r}, \theta)$.
10. Define a proper non-dimensional form for the volume flux $\tilde{Q}$ and obtain it by integration of the velocity profile.

$$1. Q(R, \kappa, \epsilon, \rho, \mu, \Delta p) \longrightarrow \frac{Q\mu}{\Delta p R^3} = f\left(\kappa, \epsilon, \frac{\rho \Delta p R^2}{\mu^2}\right)$$

$$2. \quad \vec{v} = v_\theta(r, \theta) \bar{e}_\theta \quad v_r = v_\phi = 0 \quad \frac{\partial}{\partial \phi} = 0$$

$$\text{cont.} \quad \frac{\partial}{\partial \theta} (v_\theta \sin \theta) = 0$$

$$\text{cur} \quad -\rho \frac{v_\theta^2}{r} = -\frac{\partial p}{\partial r}$$

$$\text{cm}\theta \quad \rho \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} = -\frac{1}{r} \frac{\partial p'}{\partial \theta} + \mu \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right)$$

$$\text{cm}\phi \quad 0 = \frac{\partial p'}{\partial \phi}$$

$$\text{domain} \quad \begin{cases} \kappa R \leq r \leq R \\ 0 \leq \theta \leq \pi \end{cases}$$

$$p' \equiv p - p_0$$

$$dp' = dp$$

$$\text{bc.} \quad v_\theta(r = \kappa R, \theta) = 0$$

$$v_\theta(r = R, \theta) = 0$$

$$p'(r = R, 0 \leq \theta \leq \epsilon) = p_i - p_0 = \Delta p$$

$$p'(r = R, \pi - \epsilon \leq \theta \leq \pi) = p_0 - p_0 = 0$$

$$Q = \int_{r=\kappa R}^{r=R} v_\theta(r, \theta = \frac{\pi}{2}) 2\pi r dr$$

$$Q = Q(R, \kappa, \epsilon, \rho, \mu, \Delta p) \quad \checkmark$$

$$4. \quad \text{cont.} \quad \frac{\partial}{\partial \theta} (v_\theta \sin \theta) = 0$$

$$\text{cm}\theta \quad 0 = -\frac{1}{r} \frac{dp'}{d\theta} + \mu \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right)$$

$$\text{orders of magnit.} \quad \frac{\Delta p}{R} \sim \frac{\mu V_\theta}{R^2} \longrightarrow V_\theta \sim \frac{\Delta p R}{\mu}$$

$$5. \quad \tilde{v}_\theta = \frac{v_\theta}{v_0} = \frac{v_0 \mu}{\Delta p R} \quad \tilde{r} = \frac{r}{R} \quad \tilde{p} = \frac{p - p_0}{\Delta p}$$

$$\text{cont.} \quad \frac{\partial}{\partial \theta} (\tilde{v}_\theta \sin \theta) = 0$$

$$\text{cmo} \quad 0 = - \frac{d\tilde{p}}{d\theta} + \frac{1}{\tilde{r}} \frac{\partial}{\partial \tilde{r}} \left(\tilde{r}^2 \frac{\partial \tilde{v}_\theta}{\partial \tilde{r}} \right)$$

$$\tilde{v}_\theta(\tilde{r} = \kappa, \theta) = \tilde{v}_\theta(\tilde{r} = 1, \theta) = 0$$

$$\tilde{p}(0 \leq \theta \leq \epsilon) = 1$$

$$\tilde{p}(\pi - \epsilon \leq \theta \leq \pi) = 0$$

$$\tilde{Q} = \frac{Q\mu}{\Delta p R^3} = 2\pi \int_{\tilde{r}=\kappa}^{\tilde{r}=1} \tilde{v}_\theta(\tilde{r}, \theta = \frac{\pi}{2}) \tilde{r} d\tilde{r}$$

$$\tilde{Q} = \tilde{Q}(\kappa, \epsilon) \quad \checkmark$$

$$6. \quad \frac{\partial}{\partial \theta} (\tilde{v}_\theta \sin \theta) = 0 \quad \longrightarrow \quad \tilde{v}_\theta \sin \theta = \tilde{u}(\tilde{r})$$

$$7. \quad \frac{d\tilde{p}}{d\theta} = \frac{1}{\tilde{r} \sin \theta} \frac{d}{d\tilde{r}} \left(\tilde{r}^2 \frac{d\tilde{u}}{d\tilde{r}} \right)$$

$$\underbrace{\sin \theta \frac{d\tilde{p}}{d\theta}}_{\text{only } f(\theta)} = \underbrace{\frac{1}{\tilde{r}} \frac{d}{d\tilde{r}} \left(\tilde{r}^2 \frac{d\tilde{u}}{d\tilde{r}} \right)}_{\text{only } f(\tilde{r})} \implies \text{constant.}$$

$$\sin \theta \frac{d\tilde{p}}{d\theta} = C$$

$$\frac{1}{\tilde{r}} \frac{d}{d\tilde{r}} \left(\tilde{r}^2 \frac{d\tilde{u}}{d\tilde{r}} \right) = C$$

C may be a function of the governing parameter κ, ϵ .

8.

$$\sec \theta \frac{d\tilde{p}}{d\theta} = C$$

$$\frac{d\tilde{p}}{d\theta} = \frac{C}{\sec \theta}$$

$$\tilde{p}(\theta = \epsilon) = 1$$

$$\tilde{p}(\theta = \pi - \epsilon) = 0$$

$$\int_1^0 d\tilde{p} = C \int_{\epsilon}^{\pi-\epsilon} \frac{d\theta}{\sec \theta} \longrightarrow C = \frac{-1}{\int_{\epsilon}^{\pi-\epsilon} \frac{d\theta}{\sec \theta}} = \frac{-1}{E(\epsilon)}$$

say $E(\epsilon)$

$$\text{with } E(\epsilon) = \int_{\epsilon}^{\pi-\epsilon} \frac{d\theta}{\sec \theta} = \left[\ln \left(\tan \frac{\theta}{2} \right) \right]_{\epsilon}^{\pi-\epsilon} = \dots = -\ln \left(\frac{1 - \cos \epsilon}{1 + \cos \epsilon} \right)$$

9.

$$\frac{1}{\tilde{r}} \frac{d}{d\tilde{r}} \left(\tilde{r}^2 \frac{d\tilde{u}}{d\tilde{r}} \right) = C$$

$$\tilde{u}(\tilde{r} = k) = 0$$

$$\tilde{u}(\tilde{r} = 1) = 0$$

$$\longrightarrow \tilde{u}(\tilde{r}) = -\frac{C}{2} \left[1 - \tilde{r} + k \left(1 - \frac{1}{\tilde{r}} \right) \right]$$

$$\tilde{u}_{\theta}(\tilde{r}, \theta) = \frac{1}{2 E(\epsilon) \sec \theta} \left[1 - \tilde{r} + k \left(1 - \frac{1}{\tilde{r}} \right) \right]$$

10.

$$\tilde{Q} = 2\pi \int_{\tilde{r}=k}^{\tilde{r}=1} \tilde{u}(\tilde{r}) \tilde{r} d\tilde{r} = \frac{\pi}{6 E(\epsilon)} (1-k)^3$$